Projective Reed–Solomon syndromes beyond redundancy four: complete first levels and coherent polar flags

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Abstract

We classify, up to projective-semilinear equivalence, the one-column maximum-distance-separable extensions of projective Reed–Solomon codes at redundancy five over every prime power $q \geq 7$, and give exact projective counts. Equivalently, we classify the deep-hole syndrome directions in the first case left open by the classification through redundancy four. A syndrome determines a Hankel linear system of binary forms and is deep exactly when that system contains no completely split squarefree member. The generic redundancy-five case is controlled by an integral off-diagonal fiber-square curve of arithmetic genus one.

For higher redundancy we introduce coherent polar flags: iterated contractions in which every selected factor remains a marked forbidden root. Their basic identity is $g \in W_{\iota_\lambda f} \Rightarrow \lambda g \in W_f$, with a squarefree lift exactly when $\lambda \nmid g$. Separating contained components from transverse point counts gives complete all-field syndrome classifications at redundancies six and seven; the redundancy-seven split-free list is a deep-hole classification whenever the covering radius is six, in particular for $q \geq 11$. More generally, products of retained markers are dense in the full binary-form parameter space, so their bottom contractions close to a projectivized catalecticant row space. Combining this reduction with an integral Grassmannian component calculation proves that the recursively pointed contained bad scheme has only persistent and modular components at every redundancy. Consequently the transverse branch has the unconditional linear-scale threshold $q \geq 6r - 15 + \lfloor 2\sqrt{6r-17} \rfloor$.

Index Terms—Algebraic coding theory, deep holes, finite fields, maximum-distance-separable extensions, normal rational curves, projective Reed–Solomon codes.

1 Introduction

This paper has three principal contributions. First, it gives the complete all-field classification at redundancy five, the first case left open by the known classification through redundancy four. Second, it introduces coherent polar flags: a marked contraction method that propagates splitting information to higher redundancies without forgetting the factor needed for a squarefree lift. This gives complete all-field syndrome classifications at redundancies six and seven. Third, a catalecticant-rowspace reduction and an integral bottom-component calculation prove that every recursively pointed contained component is persistent or modular. We use deep-hole syndromes because their Hankel systems make the base classification, propagation mechanism, and contained-component geometry visible.

The mechanism is the elementary identity

$$g \in W_{\iota_\lambda f} \iff \lambda g \in W_f, \qquad \lambda g \text{ squarefree} \iff \lambda \nmid g$$

when g is squarefree. Retaining λ is therefore the essential coherence condition. Theorem 1.3 classifies the contained components, and Theorem 5.13 closes the transverse branch for

$$q \geq 6r - 15 + \left\lfloor 2\sqrt{6r - 17} \right\rfloor .$$

Let $\mathrm{PRS}(k)$ denote the projective Reed–Solomon code of length $q+1$ obtained by evaluating homogeneous binary forms of degree $k-1$ on $\mathbb{P}^1(\mathbb{F}_q)$. If the redundancy is

$$r = q + 1 - k,$$

a parity-check matrix has as columns the $\mathbb{F}_q$-points of the degree-$(r-1)$ normal rational curve

$$\mathcal{R}_{r-1} = \{(1, t, \ldots, t^{r-1}) : t \in \mathbb{F}_q\} \cup \{(0, \ldots, 0, 1)\} \subset \mathbb{P}^{r-1}.$$

Call a projective syndrome direction *split-free* if it is not contained in the span of $r-2$ distinct columns. When $\rho(\mathrm{PRS}(q+1-r)) = r-1$, the split-free directions are precisely the projective syndrome directions of deep-hole cosets. The logical separation used throughout the paper is

$$\boxed{\text{split-free classification} \ + \ \rho(\mathrm{PRS}(q+1-r)) = r-1 \quad \Longrightarrow \quad \text{deep-hole classification.}}$$

The first term is proved geometrically; the covering-radius equality is a separate coding-theoretic input and is never inferred from a finite split-free census.

The correspondence between deep holes, one-symbol MDS extensions, and uncovered points of a normal rational curve was developed in [Kai17, ZWK20]. Zhang, Wan, and Kaipa classified projective Reed–Solomon deep-hole directions through redundancy four and identified redundancy five as the next case [ZWK20]. Our first theorem gives the corresponding all-field redundancy-five classification.

Theorem 1.1 (Complete redundancy-five classification)**.** Let $q \geq 7$ be a prime power. The covering radius of $\mathrm{PRS}(q-4)$ is 4. Its projective deep-hole syndromes are exactly the following disjoint families:

(i) the tangent family T, of size $q(q+1)$;

(ii) the conjugate-secant family S, of size $q(q^2-1)/2$;

(iii) if $\operatorname{char} \mathbb{F}_q \neq 3$, the rational osculating-pair family O^+ when $q \equiv 2 \pmod 3$, of size $q(q+1)/2$, or the conjugate osculating-pair family O^- when $q \equiv 1 \pmod 3$, of size $q(q-1)/2$;

(iv) in characteristic three, one $\mathrm{PGL}_2(q)$-fixed nucleus point and one wild Artin–Schreier orbit W of size $(q^2-1)/2$;

(v) the sporadic $\mathrm{PGL}_2(q)$-orbits in Table 4, occurring exactly for $q \in \{7, 8, 9, 11, 13, 17, 19\}$.

There are no other deep-hole syndromes. The three corresponding nonsporadic totals are

$$\frac{q^2(q+3)}{2}, \qquad \frac{q(q+1)(q+2)}{2}, \qquad \frac{q^3+3q^2+q+1}{2},$$

according as $q \equiv 1 \pmod 3$ with characteristic not three, $q \equiv 2 \pmod 3$, or $\operatorname{char} \mathbb{F}_q = 3$.

The main mechanism is a characteristic-free Hankel dictionary. A syndrome at redundancy r annihilates a linear system of binary forms of degree $r-2$. It is split-free if and only if this system contains no completely split squarefree form; it is a deep-hole direction when, in addition, $\rho(\mathrm{PRS}(q+1-r)) = r-1$. At redundancy five the system is a pencil of cubics. Its induced degree-three map of projective lines is either cyclic or has geometric monodromy S_3.

Cyclic descent gives the arithmetic families in Theorem 1.1; in the S_3 case the off-diagonal fiber square is an integral curve of arithmetic genus one, and the singular-curve Weil bound [AP95] forces a split fiber for $q \geq 23$. The certified finite computations in Section 7 handle the remaining fields.

Starting at redundancy six, the Hankel system is no longer a pencil. The useful replacement is the coherently parameterized line of contractions obtained by choosing one prospective root. Iteration produces a *polar flag*. Geometrically, naïve contraction keeps the lower divisor but forgets which point of $\mathbb{P}^1$ was removed. If the lower divisor passes through that point, putting the point back creates a double point rather than a squarefree divisor. The polar flag retains the removed point as a forbidden marker, exactly the datum needed for lifting. This pointed coherence is essential: at $q = 19$ an individual contracted fiber has no split witness avoiding its marker, but no consecutive-row polar line is contained in its orbit.

The resulting structural theorem has two outputs.

Theorem 1.2 (Classification results and explicit gates beyond five)**.** The following statements hold.

(a) For every prime power $q \geq 7$, the projective deep-hole directions of $\mathrm{PRS}(q-5)$ are exactly the persistent tangent and conjugate-secant families, the exceptional orbits listed for $q = 7, 8, 9, 11, 13$, and, when $q = 2^m$ with odd $m \geq 5$, one characteristic-two nucleus orbit. Here $m \geq 5$ avoids double counting: $m = 3$, or $q = 8$, is already the $|\mathcal{O}| = 9$ row of the exceptional table.

(b) The displayed finite domains through $q = 32$ have the certified split-free classification stated in Section 6. Together with the structural theorem, these finite classifications give the all-field result: for every $q \geq 13$ the split-free locus consists of the persistent families and, when $q = 2^m$ with m odd, one central nucleus point. This is a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$, in particular for $q \geq 11$.

At redundancy r, the semilinear orbits in the persistent conjugate-secant family are parameterized by

$$T/T^{r-1} \quad \text{modulo inversion and Frobenius},$$

where $T = \{z \in \mathbb{F}_{q^2}^{\times} : z^{q+1} = 1\}$ is the norm-one torus and Frobenius is coefficient Frobenius. On the tangent family the unipotent coordinate transforms as $z \mapsto z + (r-1)u$; consequently its orbit decomposition changes precisely when $\operatorname{char} \mathbb{F}_q$ divides $r - 1$.

The two parts have different scopes. Both are unconditional all-field syndrome classifications, but part (b) becomes a deep-hole classification only when $\rho(\mathrm{PRS}(q-6)) = 6$. Table 2 places the split-free result, radius gate, deep-hole conclusion, and remaining field gap in separate columns for each redundancy.

Theorem 1.3 (Stable components and the uniform transverse consequence)**.** The stable-component assertion $\mathrm{SC}(j)$ holds for every $j \geq 6$. In characteristic two, the modular component descended from the R5 cyclic plane is empty for every $r \geq 8$. Consequently, if

$$q \geq 6r - 15 + \left\lfloor 2\sqrt{6r-17} \right\rfloor,$$

every split-free redundancy-r syndrome belongs to $\mathcal{P}_r \cup \mathcal{M}_r$. Calling these directions code deep holes still requires the separate covering-radius equality $\rho(\mathrm{PRS}(q+1-r)) = r - 1$.

Theorem 1.3 is the third principal contribution. Its proof is given in Section 5: products of the retained markers are dense in the full binary-form parameter space, so the bottom contractions close to the projectivized row space of a consecutive catalecticant. Irreducibility reduces containment to one bottom component, and each component is then treated explicitly.

Relation to previous work. The tangent and conjugate-secant families and the coding/normal-rational curve dictionary come from [Kai17, ZWK20]; the covering-radius input in the high-rate range is due to Seroussi and Roth [SR86]. The broader arc/MDS and code-automorphism background is surveyed or established in [BL19, Dür87]. Earlier explicit-family results for projective and generalized projective Reed–Solomon codes include [ZW17, XHX17, Xu23]; the recent extension framework of [WDC23] further clarifies the deep-hole/MDS-extension interface. These works construct families, treat punctured or lower-redundancy regimes, or analyze the extension criterion; they do not supply the complete redundancy-five cubic-pencil exhaustion proved here. Classical apolarity, catalecticants, and binary-form rank strata are treated in [IK99, CS11]; our use of divided powers keeps the small-characteristic contraction formulas explicit. From the viewpoint of binary forms, the question is which linear systems contain a totally split squarefree member. Factorization statistics in linear families [CMP17] and normal-rational-curve nuclei [GH00] are inputs. For splitting families, we use Wang's Theorem 3.6 [Wan26]: on the étale locus, the factorization type of a specialized polynomial is the cycle type of its Frobenius conjugacy class. The coding-theory references above identify deep syndromes with uncovered points of a normal rational curve and settle the lower-redundancy families; they do not classify the redundancy-five cubic pencils. The orbit-theoretic and factorization references supply normal forms and distribution tools, but not the coherent marked contraction used here. The nucleus results locate characteristic-dependent kernels, while the present work computes which of their coherent lifts meet the PRS splitting incidence. Accordingly, this paper's contributions are separated into the redundancy-five through seven classifications and the conditional transverse induction mechanism. Claim-by-claim novelty and dependency boundaries are recorded in the electronic ledger.

Organization. Section 2 first gives a proof-free guide to the argument, and Section 3 gives the Hankel dictionary. Section 4 proves Theorem 1.1. Sections 5 and 6 develop coherent polar induction and prove Theorem 1.2. Section 7 states the computational and formal trust boundary, and Section 8 records the exact open boundary.

2 Guide to the proof

This section is a proof-free guided tour. Its purpose is to install the information flow used throughout the paper before coordinates, finite exceptional tables, or formal trust boundaries intervene.

Syndrome to splitting system. A syndrome direction $[f] \in \mathbb{P}(\Gamma^{r-1}E)$ determines its Hankel annihilator $W_f \subset \operatorname{Sym}^{r-2} E^\vee$. A product of $r-2$ distinct rational linear factors lies in W_f exactly when f lies in the span of the corresponding normal-rational-curve columns. Thus the geometric classification begins with split-free systems. The separate covering-radius theorem is what promotes split-free directions to code deep holes; the dictionary alone does not.

The redundancy-five base cover. At redundancy five, W_f is a pencil of binary cubics. Its root incidence is a degree-three map of projective lines. The off-diagonal fiber square separates the cyclic cases, controlled by Frobenius on a deck transformation, from the S_3 case, controlled by rational points on a geometrically integral $(2, 2)$ curve. This is the bottom cover used at every later level.

Why independent lower fibers fail. At higher redundancy, choosing one lower witness forgets which factor was removed. If that lower witness contains the chosen factor, then multiplication restores it with multiplicity two rather than producing a squarefree form. The lost datum is therefore not auxiliary: it is the exact obstruction to lifting.

Coherent replacement and the fork. Choose a prospective root λ, contract f to $\iota_\lambda f$, and retain λ as a marked forbidden root. Iteration gives an ordered polar flag. Its first-polar line has two qualitatively different behaviors. If it is transverse to the lower bad locus and collision schemes, only finitely many parameters must be removed and a point count produces an escape. If it is scheme-theoretically contained, the point-count argument says nothing: ordinary catalecticant components and components created in small characteristic require a separate level-specific classification. Section 5 introduces the persistent and modular loci only when that distinction enters the proof.

Integral component closure. The ordered-root incidence of the bottom cubic pencil is a symmetric $(2,2)$ form on a Grassmannian. Unique factorization leaves collision, one cyclic branch, and in characteristic three an anti-invariant diagonal collision. An integral Plücker-to-syndrome bridge identifies the cyclic branch with the coherent-Fano carrier, meaning the coefficient scheme on which the entire consecutive polar line lies in that carrier. After persistent saturation that carrier is empty, with characteristics two and three treated before residualization.

The all-level step is not formal marker substitution. Products of the retained markers are dense in the full binary-form parameter space, so their bottom contractions close to

$$\mathbb{P}(\operatorname{rowspace}\operatorname{Cat}_{m,4}(f))\,.$$

This irreducible projective space lies in one component of the finite bottom bad union. The rank-two component is handled by the integral consecutive-catalecticant theorem, the tame Veronese contains no line, the characteristic-three cone is handled by its explicit ruling calculation, and the characteristic-two plane gives two overlapping zero blocks. Those blocks cover every coefficient from redundancy eight onward.

Lifting. On the transverse branch, a rational point of the lower ordered splitting cover outside the branch, diagonal, marker, and collision divisors produces a split squarefree lower witness avoiding all markers. Multiplying by the marked factors then lifts it squarefreely to W_f. Figure 1 records this sequence at the point where the construction is formalized.

Table 1: Mathematical dependency map.

Result	Printed geometric argument	Imported input
R5	Hankel dictionary; gcd strata; cyclic descent; integral $(2,2)$ fiber square and point bound	lower tangent/secant families; high-rate radius theorem; singular-curve bound
R6	one-marker polar line; ordinary and binary contained components; transverse escape	R5 lower package and radius gate
R7	two-marker contraction; complete second-marker bad scheme, including cyclic and fixed-gcd collisions; contained rank-two theorem; transverse escape	R5/R6 lower packages; separate covering-radius result
All $r \geq 6$	catalecticant-rowspace transport; integral ordered-root incidence and factorization; Plücker bridge; coherent-Fano elimination; separate vertical fibres	R5 bottom cover and persistent/Lucas pullback identities

Table 2: Classification status by redundancy. "Unconditional" refers to the displayed field range. A split-free classification becomes a deep-hole classification only after the separate radius gate.

Red.	Field range	Split-free geometry	Deep holes/MDS extensions	Remaining gap
5	$q \geq 7$	Complete, unconditional	Complete, unconditional; $\rho = 4$	None
6	$q \geq 7$	Complete, unconditional	Complete, unconditional; $\rho = 5$	None
7	$q \geq 7$	Complete, unconditional	Complete for $q \geq 11$; conditional on $\rho = 6$ at $q = 7, 8, 9$	Radius at $q = 7, 8, 9$
$r \geq 6$	$q \geq Q_r$	Contained in $\mathcal{P}_r \cup \mathcal{M}_r$, unconditional	Conditional on radius and exact modular membership	Fields below Q_r; radius; modular arithmetic

3 The syndrome–Hankel dictionary

Let E be a two-dimensional vector space. We use divided powers for syndromes and ordinary symmetric powers for root forms. This avoids invalid binomial rescalings in small characteristic. For $f = (a_0 : \cdots : a_{r-1}) \in \mathbb{P}(\Gamma^{r-1}E)$, write $e_0, \ldots, e_{r-1}$ for the divided-power coordinate basis and $\nu(t) = (1, t, \ldots, t^{r-1})$, with $\nu(\infty) = e_{r-1}$. If x, y is a basis of E with dual basis X, Y, our perfect divided-power/symmetric-power pairing is

$$\left\langle x^{[n-i]}y^{[i]}, X^{n-j}Y^j \right\rangle = \delta_{ij}, \qquad \left\langle \sum_{i=0}^{n} a_i x^{[n-i]}y^{[i]}, \sum_{i=0}^{n} b_i X^{n-i}Y^i \right\rangle = \sum_{i=0}^{n} a_i b_i.$$

Thus the pairing is coefficient extraction, with no binomial factors in any characteristic. Define

$$H_f = \begin{pmatrix} a_0 & a_1 & \cdots & a_{r-2} \\ a_1 & a_2 & \cdots & a_{r-1} \end{pmatrix}, \qquad W_f = \ker H_f \subset \operatorname{Sym}^{r-2} E^\vee.$$

Table 3: Notation used from the Hankel dictionary onward.

Symbol	Meaning
q	field size; all numerical thresholds are thresholds on prime powers
k	code dimension in $\mathrm{PRS}(k)$
$r = q + 1 - k$	redundancy
$n = r - 1$	divided-power degree of a syndrome representative
E	a two-dimensional vector space over the current base field
$f \in \Gamma^n E$	divided-power representative of a projective syndrome
$W_f \subset \operatorname{Sym}^{n-1} E^\vee$	Hankel/apolar witness system annihilated by f
g, δ, κ	genus, deletion degree, and point-bound correction
d	degree of the unavailable marker scheme

Read $f = (a_0, \ldots, a_{r-1})$ as a finite sequence. A form $g \in W_f$ is the characteristic polynomial of a recurrence satisfied by that sequence, while a squarefree split g has a basis of geometric-sequence solutions. Thus split-free means that the syndrome sequence satisfies no order-$(r-2)$ recurrence with squarefree completely split characteristic polynomial.

Lemma 3.1 (Hankel criterion)**.** Let $\lambda_1, \ldots, \lambda_{r-2}$ be distinct points of $\mathbb{P}(E^\vee)(\mathbb{F}_q)$ and put $g = \prod_i \lambda_i$. Then

$$f \in \operatorname{span}\big(\nu(\lambda_1), \ldots, \nu(\lambda_{r-2})\big) \iff g \in W_f.$$

The same statement with repeated factors and Hasse derivatives gives the corresponding osculating spans. Both statements commute with $\mathrm{P\Gamma L}_2(q)$.

Proof. On an affine chart, the geometric sequences $\nu(\lambda_i)$ solve the recurrence with characteristic polynomial g; their independence is the Vandermonde determinant. A root at infinity gives the reversed-chart recurrence. For repeated roots, the confluent Vandermonde basis is formed by Hasse derivatives, so the same argument works in every characteristic. Equivariance follows because substitution transports both the normal rational curve and the recurrence. □

Corollary 3.2 (Split-free criterion)**.** Assume $\rho(\mathrm{PRS}(q+1-r)) = r-1$. A projective syndrome f is deep if and only if W_f contains no completely split squarefree form of degree $r-2$ over $\mathbb{F}_q$.

We use *split-free* without a covering-radius hypothesis and call a direction *shallow* when W_f contains a completely split squarefree form.

The first locus that persists under contraction is already visible here. If every member of W_f has a common quadratic factor, then that factor is either a rational square, an irreducible quadratic, or a product of two distinct rational factors. These are respectively the tangent deep family, the conjugate-secant deep family, and a shallow rational secant. The common factor cannot grow with redundancy: since $\dim W_f = r-3$,

$$r - 3 \leq r - 1 - \deg \gcd(W_f),$$

so its degree is at most two.

3.1 Coding consequences and recognition

The dictionary has two immediate coding consequences. First, a split-free direction is an uncovered point of the normal rational curve and hence gives a one-column MDS extension whenever the stated covering-radius gate applies. The classification theorems therefore give exact projective deep-hole counts, not only normal forms. Second, they give recognition procedures. From a syndrome one forms H_f, computes W_f, and tests the listed geometric families: a common quadratic factor detects the persistent tangent and conjugate-secant families; the relevant contraction kernel detects the modular locus; and the finite exceptional fields are decided by the canonical orbit representatives printed in Table 4 and recorded in the supplement.

Recognition at the fixed redundancies proved here begins with row-reduction of H_f and the listed family-membership tests. Coherent polar induction replaces a direct search through the projective members of W_f.

4 Redundancy five: exceptional cubic covers

Here $f = (a_0 : \cdots : a_4)$ and W_f is the pencil annihilated by

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \end{pmatrix}.$$

These are divided-power coordinates; the coefficients of a cubic in W_f are ordinary symmetric-power coordinates. Factoring the formal quartic $\sum a_i T^i U^{4-i}$ would therefore be noninvariant in characteristics two and three.

Proposition 4.1 (Radius gate)**.** For every prime power $q \geq 7$,

$$\rho(\mathrm{PRS}(q-4)) = 4.$$

Proof. The coding/arc criterion identifies radius four with completeness of the quartic normal rational curve. Seroussi–Roth prove completeness when

$$k \geq \left\lfloor \frac{q-1}{2} \right\rfloor$$

apart from the even-field dimensions $k = 2, q-2$ [SR86]. Here $k = q-4$, the inequality holds for $q \geq 7$, and neither exceptional dimension occurs. Thus Corollary 3.2 applies. The finite certificate independently checks the same conclusion only in its declared finite domain; it is not the all-field radius proof. □

4.1 Gcd strata

Proposition 4.2 (Quadratic gcd)**.** If $\deg \gcd(W_f) = 2$, then

$$W_f = Q\langle T, U \rangle.$$

If Q is split squarefree, f is shallow. If Q is irreducible, f belongs to the conjugate-secant family; if $Q = \lambda^2$, it belongs to the tangent family. Conversely every point in those families has the displayed gcd. Their deep-family sizes are

$$|\mathrm{T}| = q(q+1), \qquad |\mathrm{S}| = \frac{q(q^2-1)}{2}.$$

Proof. The equality follows by dimension. A split Q and a linear cofactor avoiding its two roots give a split squarefree cubic. An irreducible Q survives in every member, while a square Q forces a repeated root in every member. Lemma 3.1, including its confluent and conjugate-root forms, identifies the corresponding syndromes with the secant and tangent spans. The sizes are the established tangent/conjugate-secant counts from the lower-redundancy classification [Kai17, ZWK20]. □

Proposition 4.3 (Linear gcd)**.** Suppose $\deg \gcd(W_f) = 1$. Then f is shallow for $q \geq 8$. At $q = 7$ the same conclusion is part of Certificate R5.

Proof. Write $W_f = \lambda_0 V$ with V a basepoint-free quadratic pencil. If its degree-two map $\psi : \mathbb{P}^1 \to \mathbb{P}^1$ is separable, its rational deck involution ι gives the graph

$$\Gamma_\iota(\mathbb{F}_q) = \{(t, \iota(t)) : t \in \mathbb{P}^1(\mathbb{F}_q)\},$$

which has exactly $q+1$ rational points. A safe union bound deletes at most two fixed graph points, at most four graph points over the branch fibers, and at most the two graph points $(\lambda_0, \iota(\lambda_0))$ and $(\iota(\lambda_0), \lambda_0)$. Thus fewer than $q+1$ graph points are deleted when $q \geq 8$. A surviving point gives a split quadratic with two distinct roots avoiding λ_0, and hence a split squarefree cubic in W_f. In characteristic two an inseparable quadratic pencil is equivalent to $\langle T^2, U^2 \rangle$; the two overlapping Hankel equations then force f onto the normal rational curve, contrary to the rank-two pencil hypothesis. □

4.2 The trivial-gcd cubic cover

Choose a basis c_1, c_2 of a trivial-gcd pencil and put

$$X_f = \{([u:v], t) : uc_1(t) + vc_2(t) = 0\} \subset \mathbb{P}^1 \times \mathbb{P}^1.$$

Proposition 4.4 (Cubic incidence and off-diagonal fiber square)**.** The curve X_f is geometrically integral of bidegree $(1,3)$ and arithmetic genus zero. Projection to the root coordinate is birational, while projection to the pencil coordinate gives a degree-three morphism

$$\phi_f : \mathbb{P}^1 \longrightarrow \mathbb{P}^1.$$

If ϕ_f is separable, division of

$$c_1(t_1)c_2(t_2) - c_2(t_1)c_1(t_2)$$

by the diagonal bracket defines a residual curve $Y_f \subset \mathbb{P}^1 \times \mathbb{P}^1$ of bidegree $(2,2)$ and arithmetic genus one. Its geometric components are the geometric-monodromy orbits on ordered pairs of distinct roots.

Proof. A component of X_f of bidegree $(0,e)$ would be a common factor of c_1, c_2; hence none exists. The bidegree genus formula gives genus zero. A root determines the unique pencil member containing it, so the root projection is birational. For a separable map the diagonal factor in the fiber-square determinant has multiplicity one. Removing its bidegree $(1,1)$ from the fiber square leaves bidegree $(2,2)$, whose arithmetic genus is one. The standard fiber-square/monodromy correspondence identifies its components with orbits on distinct ordered sheets. $\square$

Lemma 4.5 (Cyclic stratum)**.** Suppose ϕ_f is geometrically cyclic.

(i) In characteristic different from three there are two ramification points. If they are rational, the corresponding osculating-plane intersection is deep exactly when $q \equiv 2 \pmod 3$; if they are conjugate, it is deep exactly when $q \equiv 1 \pmod 3$.

(ii) In characteristic three all osculating planes meet in $n = (0:0:1:0:0)$, whose pencil is $\langle T^3, U^3 \rangle$; this fixed point is deep.

(iii) The remaining wild characteristic-three pencils have the form $\langle U^3, T^3 + aTU^2 \rangle$. They are deep exactly when $-a$ is nonsquare, forming one orbit of size $(q^2-1)/2$.

Proof. Assume first that the characteristic is not three. Riemann–Hurwitz gives two totally ramified points t_1, t_2. The confluent Hankel criterion places f in $\mathrm{Osc}(t_1) \cap \mathrm{Osc}(t_2)$; after transporting the pair to $\{0, \infty\}$, the pencil is $\langle T^3, U^3 \rangle$ and a geometric deck generator is $t \mapsto \zeta t$.

For a rational ramification pair, Frobenius fixes the points and the deck map descends exactly when $\zeta^q = \zeta$, namely $q \equiv 1 \pmod 3$. Thus its complement is deep exactly for $q \equiv 2 \pmod 3$. The setwise stabilizer of the ordered normal form is the split dihedral group of order $2(q-1)$, so the orbit size is

$$\frac{q(q^2-1)}{2(q-1)} = \frac{q(q+1)}{2}.$$

For a conjugate pair Frobenius interchanges the fixed points and inverts the multiplier; descent is instead $q \equiv 2 \pmod 3$. The nonsplit dihedral stabilizer has order $2(q+1)$ and the orbit size is $q(q-1)/2$.

Now suppose the characteristic is three. The second Hasse derivative of

$$\nu(t) = (1, t, t^2, t^3, t^4)$$

is $(0,0,1,3t,6t^2) = e_2$, including in the reversed infinity chart. Thus every osculating plane contains $n = [e_2]$. Substitution shows that n is $\mathrm{PGL}_2(q)$-fixed and $W_n = \langle T^3, U^3 \rangle$, so it is deep.

A remaining separable cyclic cover has one wild totally ramified point. Sending it to infinity and comparing coefficients under a nontrivial translation gives

$$W_f = \langle U^3, T^3 + \alpha T^2U + \beta TU^2\rangle, \qquad \alpha = 0, \quad \kappa^2 = -\beta.$$

It is therefore represented by

$$f_a = e_2 - ae_4, \qquad W_{f_a} = \langle U^3, T^3 + aTU^2\rangle.$$

The affine members are $z^3 + az + s$. Such a member has three rational roots exactly when the additive map $z \mapsto z^3 + az$ has a nonzero rational kernel, equivalently when $-a$ is a square. Hence the nonsquare parameters form the deep orbit. Its stabilizer is $t \mapsto \pm t + \beta$, of order $2q$, and orbit–stabilizer gives $(q^2 - 1)/2$. □

Lemma 4.6 (S_3 stratum)**.** If ϕ_f has geometric monodromy S_3 and $q \geq 23$, then the pencil contains a completely split squarefree cubic.

Proof. By Proposition 4.4, components of Y_f correspond to monodromy orbits on distinct ordered roots. The S_3 action is two-transitive, so Y_f is geometrically integral and has arithmetic genus one. The Aubry–Perret singular-curve bound, in the form

$$\left|\#C(\mathbb{F}_q) - (q + 1)\right| \leq 2\pi_C\sqrt{q}$$

for a reduced geometrically integral projective curve of arithmetic genus π_C, gives

$$\#Y_f(\mathbb{F}_q) \geq q + 1 - 2\sqrt{q}$$

[AP95, p. 468]. An integral curve of arithmetic genus one has at most one geometric singular point: if it is singular, its normalization has genus zero and the total delta invariant is one. We therefore delete at most one rational singular point. The diagonal costs

$$Y_f \cdot \Delta = (2, 2) \cdot (1, 1) = 4.$$

Riemann–Hurwitz gives different degree four for the separable cubic map, hence at most four branch values. A singular cubic has at most two distinct roots and at most two ordered off-diagonal pairs, so the branch deletion costs at most eight. If $q + 1 - 2\sqrt{q} > 13$, a smooth rational point remains outside the diagonal and branch fibers. It gives two distinct rational roots of a squarefree cubic, whose third root is rational. This inequality is equivalent to $q - 2\sqrt{q} > 12$ and holds for every prime power $q \geq 23$. □

Proposition 4.7 (Exhaustion and finite bridge)**.** The preceding cases exhaust every redundancy-five pencil for $q \geq 23$. Certificate R5 closes exactly

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19\}.$$

It finds sporadic S_3-stratum orbits exactly at $q = 7, 8, 9, 11, 13, 17, 19$ and none at $q = 16$.

Proof. The gcd degree is zero, one, or two. Propositions 4.2 and 4.3 settle the positive-gcd cases. In the trivial-gcd case the cubic map is inseparable or separable. The inseparable case occurs only in characteristic three and the overlap equations give the all-cube nucleus pencil. A separable transitive cubic has geometric monodromy C_3 or S_3, settled by Lemmas 4.5 and 4.6. The stated finite residue, canonical representatives, stabilizers, histograms, and Frobenius links are exactly the domain of Certificate R5. □

Table 4: Complete sporadic orbit inventory at redundancy five. A representative $[a_0, \ldots, a_4]$ is in the divided-power syndrome coordinates of Section 3. The last column records the nontrivial Frobenius fusion; a dash means the orbit is fixed.

q	representative	orbit size	stabilizer	Frobenius
7	`[0,1,0,0,3]`	28	12	–
7	`[0,1,0,0,2]`	112	3	–
7	`[0,0,1,0,3]`	168	2	–
7	`[0,1,0,3,2]`	168	2	–
7	`[1,0,0,2,1]`	168	2	–
8	`[0,1,1,0,2]`	252	2	$\to [0, 1, 1, 0, 4]$
8	`[0,1,1,0,4]`	252	2	$\to [0, 1, 1, 0, 6]$
8	`[0,1,1,0,6]`	252	2	$\to [0, 1, 1, 0, 2]$
9	`[1,0,0,1,2]`	180	4	–
9	`[0,1,0,4,1]`	360	2	$\leftrightarrow [0, 1, 0, 4, 4]$
9	`[0,1,0,4,4]`	360	2	$\leftrightarrow [0, 1, 0, 4, 1]$
11	`[1,0,0,1,10]`	330	4	–
11	`[1,0,0,1,1]`	660	2	–
13	`[0,1,0,0,4]`	182	12	–
13	`[1,0,0,4,7]`	546	4	–
17	`[1,0,0,1,5]`	1224	4	–
19	`[0,1,0,0,4]`	570	12	–

The certificate additionally records the complete cubic-pencil member histogram of each representative. Repeated orbit sizes in Table 4 are therefore visibly distinguished without requiring a working-directory file.

Table 5: Independent R5 completeness totals. The first equality is the direct syndrome enumeration versus the orbit–stabilizer sum; the second decomposes the same total into the formula in Theorem 1.1 and the sporadic rows above.

q	direct = orbit sum	nonsporadic	sporadic
7	$889 = 889$	245	644
8	$1116 = 1116$	360	756
9	$1391 = 1391$	491	900
11	$1848 = 1848$	858	990
13	$2080 = 2080$	1352	728
16	$2432 = 2432$	2432	0
17	$4131 = 4131$	2907	1224
19	$4541 = 4541$	3971	570

5 Coherent polar induction

5.1 Spaces, markers, and bad schemes

Fix a field k, a two-dimensional k-space E, and

$$S_n = \mathbb{P}(\Gamma^n E).$$

For $\lambda \in E^\vee$, define divided-power contraction by

$$\langle \iota_\lambda f, g\rangle = \langle f, \lambda g\rangle.$$

Then

$$g \in W_{\iota_\lambda f} \quad \Longleftrightarrow \quad \lambda g \in W_f. \tag{1}$$

The lower witness lifts squarefreely only when $\lambda \nmid g$. Thus the factor selected for contraction must remain as a marked forbidden root.

Definition 5.1 (Persistent and modular loci)**.** The *persistent locus* is the ordinary rank-two catalecticant locus propagated by contraction. The *modular locus* is the additional contraction-kernel locus that can appear in positive characteristic. A *collision divisor* records parameters at which the marked factor becomes a repeated root.

Definition 5.2 (Contraction and marker spaces)**.** Let $C_f : E^\vee \to \Gamma^{n-1}E$ be $\lambda \mapsto \iota_\lambda f$, and put $U_f = \mathbb{P}(E^\vee) \setminus \mathbb{P}(\ker C_f)$. The first-polar rational map is

$$\Phi_f : U_f \longrightarrow S_{n-1}, \qquad [\lambda] \longmapsto [\iota_\lambda f].$$

Its image is a point or a line. For $j \geq 1$, let

$$\mathrm{Conf}_j(\mathbb{P}(E^\vee)) = \{(\lambda_1, \ldots, \lambda_j) : \lambda_i \neq \lambda_\ell \text{ for } i \neq \ell\}.$$

The ordered marker space $U_{f,j}$ is the open subset on which every successive contraction is nonzero. The universal j-step polar flag is the map

$$(\lambda_1, \ldots, \lambda_j) \longmapsto \big([\iota_{\lambda_1} f], \ldots, [\iota_{\lambda_j} \cdots \iota_{\lambda_1} f]\big).$$

In an affine coordinate r the first map is

$$\Phi_f(r) = (a_1 - ra_0, \ldots, a_n - ra_{n-1}), \qquad \Phi_f(\infty) = (a_0, \ldots, a_{n-1}).$$

Example 5.3 (The complete R5-to-R6 propagation at $q = 29$)**.** Work over $\mathbb{F}_{29}$ at redundancy six and take

$$f = (6 : 1 : 0 : 0 : 5 : 1).$$

The Hankel kernel is the net of quartics

$$W_f = \ker \begin{pmatrix} 6 & 1 & 0 & 0 & 5 \\ 1 & 0 & 0 & 5 & 1 \end{pmatrix} = \langle (0,0,1,0,0), (24,1,0,1,0), (28,1,0,0,1) \rangle.$$

The displayed basis has no common factor, so the persistent test does not decide the syndrome. Choose the first marker $r = 0$. Then

$$\iota_0 f = (1 : 0 : 0 : 5 : 1).$$

This is now a redundancy-five syndrome, and its full cubic pencil is

$$W_{\iota_0 f} = \langle (0,1,0,0), (25,0,1,24) \rangle.$$

The member with pencil parameter $[9 : 28]$ is

$$h(t) = 4 + 9t - t^2 + 5t^3 \in W_{\iota_0 f},$$

because the two Hankel products are $4 + 5 \cdot 5 = 0$ and $5(-1) + 5 = 0$ in $\mathbb{F}_{29}$. Direct factorization gives

$$h(t) = 5(t-2)(t-6)(t-27).$$

None of these roots is the marked root 0. Equation (1) therefore lifts h to

$$t\,h(t) = (0, 4, 9, 28, 5) \in W_f,$$

a squarefree quartic with roots $0, 2, 6, 27$. The Hankel criterion now places f in the span of the corresponding four normal-rational-curve columns, so f is shallow. This one calculation exhibits every interface in the R5-to-R6 propagation: the upper Hankel net, the ordinary common-factor test, contraction to the complete R5 cubic pencil, selection on its splitting cover, marker avoidance, and squarefree lifting. In the general proof the R5 curve argument supplies such a pencil parameter after deleting the marked fibers; the displayed parameter $[9 : 28]$ is one complete instance of that step.

Definition 5.4 (One-step lower package)**.** Fix $f \in S_n(\mathbb{F}_q)$ with injective contraction map, so that $\Phi_f : \mathbb{P}(E^\vee) \to S_{n-1}$ is defined everywhere. A one-step lower package abstracts the R6 argument: a lower bad locus, the R5 ordered-root curve with its deletions, and the finite set of markers where that construction is unavailable. It consists of:

(i) a closed bad scheme $B_{n-1} \subset S_{n-1}$ and a finite stratification $S_{n-1} \setminus B_{n-1} = \coprod_\alpha X_\alpha$;

(ii) for every $\lambda \in \mathbb{P}(E^\vee)(\mathbb{F}_q)$ with $b = \iota_\lambda f \in X_\alpha(\mathbb{F}_q)$, a specified proper geometrically integral curve $Y_{\alpha,b}$, defined over $\mathbb{F}_q$, with genus bound g_α and point bound

$$\#Y_{\alpha,b}(\mathbb{F}_q) \geq q + \kappa_\alpha - 2g_\alpha\sqrt{q};$$

together with a zero-dimensional closed deletion scheme $D_{\alpha,b,\lambda} \subset Y_{\alpha,b}$, of degree at most δ_α, containing the singular, branch, diagonal, marker, and collision points, such that every rational point outside $D_{\alpha,b,\lambda}$ determines an ordered rational split squarefree $g \in W_b$ with $\lambda \nmid g$;

(iii) a finite parameter scheme $Z_f \subset \mathbb{P}(E^\vee)$, of degree at most d, containing $\Phi_f^{-1}(B_{n-1})$ and every parameter at which the preceding construction is unavailable, including parameters where a fixed factor of the lower system equals a retained marker.

The index α includes any constant-field or Frobenius twist; the assertion in (iii) is the required descent statement. All degrees are geometric degrees on the displayed curve or parameter line.

The construction is summarized in Figure 1. The upper branch is the level-specific component calculation; the lower branch is the general transverse argument.

$[f] \in S_n$ $\dashrightarrow$ $\Phi_f(U_f) \subset S_{n-1}$ $\longrightarrow$ lower ordered splitting cover

$\searrow$ contained persistent/Lucas component,

$\nearrow$ transverse delete branch, diagonal, and markers

$\downarrow$ lift by (1)

split squarefree witness in W_f.

Figure 1: Coherent polar induction. The contained branch is a level-specific component theorem; only the transverse branch is controlled by the general point-count argument.

Theorem 5.5 (Polar-flag construction)**.** The maps of Definition 5.2 commute with field extension and the natural $\mathrm{GL}(E)$ action. They include the infinity chart displayed above. If $g \in W_{\iota_{\lambda_j} \cdots \iota_{\lambda_1} f}$, then $\lambda_1 \cdots \lambda_j g \in W_f$; the lift is squarefree exactly when g is squarefree and avoids every marker λ_i.

Proof. Contraction is defined by the perfect divided-power/symmetric-power pairing, so it commutes with base change and change of basis. Repeated application of (1) gives the membership assertion. Since the λ_i are distinct on the configuration space, their product is squarefree; the product with g is squarefree precisely when g is squarefree and none of the λ_i divides it. The coordinate formula at infinity follows by evaluating C_f on the second basis covector. $\square$

Lemma 5.6 (Three readings of the collision divisor)**.** Assume W_f has trivial gcd, and let $\lambda \in \mathbb{P}(E^\vee)$. The following are equivalent:

(i) every member of W_f divisible by λ is divisible by λ^2;

(ii) λ is a fixed factor of $W_{\iota_\lambda f}$;

(iii) λ lies on the collision divisor of the linear series W_f.

Proof. Equation (1) gives the equality of subspaces

$$\lambda W_{\iota_\lambda f} = W_f \cap \lambda \operatorname{Sym}^{n-2} E^\vee.$$

Thus (i) and (ii) are the same statement after dividing by λ. Since W_f is base-point-free, (i) says exactly that the morphism defined by W_f has zero differential at λ, which is (iii). $\square$

Lemma 5.7 (Uniform collision alternatives)**.** Let $j \geq 5$, and let a base-point-free g^{j-4}_{j-2} on $\mathbb{P}^1$ define an inseparable morphism in characteristic p. Then

$$p \mid (j-2), \qquad p(j-4) \leq j-2.$$

The only possible inseparable cases are $(j,p) = (5,3)$ and $(6,2)$. In the separable case the collision divisor has degree at most $2j-6$; in particular this holds unconditionally for $j \geq 7$.

Proof. An inseparable morphism factors through absolute Frobenius. Hence $p \mid (j-2)$, and its $j-3$ sections lie in the $(j-2)/p+1$-dimensional space of p-th powers. This gives the inequality. Since $(j-2)/(j-4) < 2$ for $j \geq 7$, no prime remains; the two smaller cases follow directly. In the separable case some value–derivative minor is nonzero; as a binary Wronskian it has degree $2(j-2)-2 = 2j-6$ and contains the collision scheme. $\square$

Lemma 5.8 (Linear modular pullbacks)**.** Let $\mathcal{N} \subset S_{j-2}$ be a projective linear modular nucleus. Then $\Phi_f^{-1}(\mathcal{N})$ is empty, one point, or the whole parameter line. In the transverse case its degree is at most one.

Proof. The polar map has linear coordinates in the parameter, so the pullback ideal is generated by linear forms on $\mathbb{P}^1$. $\square$

Definition 5.9 (One-step contained-component assertion)**.** For a specified bad scheme B_{n-1}, collision divisor, and explicit closed list $\mathcal{C}_n \subset S_n$, the assertion $\mathrm{CC}(n,1;\mathcal{C}_n)$ says that every f for which the contraction map is noninjective and hence supplies no polar line, $\Phi_f^* B_{n-1}$ is the whole parameter line and hence supplies no transverse marker, or the collision divisor is the whole line and hence every lift repeats its marked root, belongs to $\mathcal{C}_n$. When the list is clear we write $\mathrm{CC}(n,1)$.

Definition 5.10 (Stable-component assertion)**.** The assertion $\mathrm{SC}(j)$ says that the recursively pointed contained bad scheme at redundancy j has no component outside $\mathcal{P}_j \cup \mathcal{M}_j$. It is a geometric component statement and contains no field-size threshold.

The indices follow two conventions. The assertion $\mathrm{CC}(n,1;\mathcal{C}_n)$ is indexed by the divided-power degree n, so $f\in S_n$ has redundancy $n+1$; $\mathrm{SC}(j)$ is indexed directly by redundancy. Accordingly, after the two base cases below, $\mathrm{SC}(j)$ is the recursive content of $\mathrm{CC}(j-1,1;\mathcal{P}_j\cup\mathcal{M}_j)$.

Proof of Theorem 1.3. We first eliminate the retained markers. Put $m=j-5$ and write

$$\mathrm{Cat}_{m,4}(f)=\begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_m & a_{m+1} & a_{m+2} & a_{m+3} & a_{m+4} \end{pmatrix}.$$

If $h=(h_0,\dots,h_m)$ is the coefficient vector of the marker product, its bottom syndrome is

$$c(h)=h\,\mathrm{Cat}_{m,4}(f). \tag{2}$$

Over an algebraic closure, products of m distinct linear forms are dense in $\mathbb{P}(\mathrm{Sym}^m E^\vee)$. The map in (2) is linear and homogeneous. After deleting its projective kernel, the scheme-theoretic closure of the attainable bottom syndromes is therefore

$$\overline{\{c(h): h \text{ is a product of distinct markers}\}}=\mathbb{P}(\operatorname{rowspace}\mathrm{Cat}_{m,4}(f))\,. \tag{3}$$

This includes the homogeneous boundary at infinity. Marker diagonals are removed only from the dense source open and reappear in its closure. Since the right-hand side is irreducible, containment in the finite bottom bad union implies containment in one of its irreducible components. We now construct and check that component ledger.

Let $\mathbf{G}=\mathrm{Gr}(2,\Gamma^3E)$ over $\operatorname{Spec}\mathbf{Z}$, and write

$$(z_0,\dots,z_5)=(p_{01},p_{02},p_{03},p_{12},p_{13},p_{23}),\qquad z_0z_5-z_1z_4+z_2z_3=0.$$

After dividing the alternating evaluation determinant of a cubic pencil by the diagonal bracket, its ordered-root incidence is

$$\begin{aligned} G_z(t,s)&=z_0+z_1(t+s)+z_2(t^2+ts+s^2)+z_3ts \\ &\quad+z_4ts(t+s)+z_5t^2s^2. \end{aligned} \tag{4}$$

Its homogenization is intrinsic on $\mathbb{P}(E^\vee)\times\mathbb{P}(E^\vee)$. We retain the closed collision incidence before taking a residual.

Over an algebraically closed field, a reducible moving part either has a vertical or horizontal factor, hence a fixed-root or collision component, or splits into two $(1,1)$ factors. Symmetry under $\tau(t,s)=(s,t)$ gives three cases. If both factors are symmetric,

$$L=a+b(t+s)+cts,\qquad M=A+B(t+s)+Cts,$$

then

$$z_0z_5-z_1z_4+z_2z_3=(ac-b^2)(AC-B^2), \tag{5}$$

so one factor has rank one and gives collision. If the factors are exchanged,

$$L=a+bt+cs+dts,\qquad G_z=L(t,s)L(s,t),$$

then

$$z_0z_5-z_1z_4+z_2z_3=(ad-bc)(ad-b^2+bc-c^2). \tag{6}$$

The first factor is again collision; the second is the cyclic branch. Finally, if $\tau L = -L$, then L is proportional to the homogeneous diagonal bracket. The product of the two anti-invariant factors has

$$(z_0, \dots, z_5) = (0, 0, -\mu, 3\mu, 0, 0)$$

and Plücker value $-3\mu^2$. It survives only in characteristic three, where it belongs to the retained diagonal collision incidence. These three cases are exhaustive by unique factorization.

For a bottom syndrome $c = (c_0, \dots, c_4)$, the kernel line of

$$H_c = \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ c_1 & c_2 & c_3 & c_4 \end{pmatrix}$$

has Plücker coordinates

$$\begin{aligned} z_0 &= c_2c_4 - c_3^2, & z_1 &= -(c_1c_4 - c_2c_3), & z_2 &= c_1c_3 - c_2^2, \\ z_3 &= c_0c_4 - c_1c_3, & z_4 &= -(c_0c_3 - c_1c_2), & z_5 &= c_0c_2 - c_1^2. \end{aligned} \tag{7}$$

Eliminating a, b, c, d from the noncollision factor of (6) gives

$$\begin{aligned} K_1 &= 3z_4^2 - 9z_2z_5 - z_3z_5, & K_2 &= z_3z_4 - 3z_1z_5, \\ K_3 &= z_3^2 - 9z_0z_5, & K_4 &= z_2z_3 - z_1z_4 + z_0z_5, \\ K_5 &= z_1z_3 - 3z_0z_4, & K_6 &= 3z_1^2 - 9z_0z_2 - z_0z_3. \end{aligned} \tag{8}$$

Thus $J = (K_1(z(c)), \dots, K_6(z(c)))$. The primitive cyclic ideal $V = (V_1, \dots, V_7)$ is

$$\begin{aligned} V_1 &= 2c_3^3 - 3c_2c_3c_4 + c_1c_4^2, \\ V_2 &= 6c_2c_3^2 - 9c_2^2c_4 + 2c_1c_3c_4 + c_0c_4^2, \\ V_3 &= 2c_1c_3^2 - 3c_1c_2c_4 + c_0c_3c_4, \\ V_4 &= c_0c_3^2 - c_1^2c_4, \\ V_5 &= 2c_1^2c_3 - 3c_0c_2c_3 + c_0c_1c_4, \\ V_6 &= 6c_1^2c_2 - 9c_0c_2^2 + 2c_0c_1c_3 + c_0^2c_4, \\ V_7 &= 2c_1^3 - 3c_0c_1c_2 + c_0^2c_3. \end{aligned} \tag{9}$$

Put

$$D = \det \begin{pmatrix} c_0 & c_1 & c_2 \\ c_1 & c_2 & c_3 \\ c_2 & c_3 & c_4 \end{pmatrix}.$$

Direct integral expansion gives

$$J \subset V, \qquad 3DV \subset J. \tag{10}$$

The supplement prints the 6×7 coefficient matrix for the second containment. Integer reduction also gives $DV \not\subset J$, and the $\mathbf{F}_3$-point $(1, 0, 0, 1, 0)$ lies on J, has $D = 2$, and does not lie on V. Hence 3 is minimal for this elimination presentation, not asserted to be an intrinsic invariant.

Over $\mathbf{Z}[1/6]$, diagonal rescaling identifies (9) with the prime homogeneous kernel of

$$[u : v : w] \longmapsto [u^2 : 2uv : v^2 + 2uw : 2vw : w^2].$$

The image point $(1, 0, 2, 0, 1)$ has $D = -6$, so $D \notin V$ and $V : D^\infty = V$. Therefore

$$(J : D^\infty) = (V : D^\infty) \text{ over } \mathbf{Z}[1/3], \qquad (J : D^\infty) = V \text{ over } \mathbf{Z}[1/6]. \tag{11}$$

The inversion of 2 in the second statement is necessary: in characteristic two, V retains a D-supported component, so $(V : D^\infty) \neq V$.

For a homogeneous carrier ideal I, define its *coherent-Fano ideal* by substituting the universal consecutive polar line into every generator of I, expanding in the two line parameters, and setting all coefficients to zero. It represents upper syndromes whose entire polar line lies in the carrier, with the required consecutive-Hankel integrability constraint.

For $d = (d_0, \dots, d_5)$, substitute $c_i = xd_i + yd_{i+1}$ in (9), and let $F_{i,k}$ be the coefficient of $x^{3-k}y^k$ in $V_i(c(x, y))$. If $H_1, \dots, H_4$ are the maximal minors, in column order, of the consecutive 3×4 Hankel matrix of d, then

$$\begin{aligned} 6H_1 &= -3F_{4,0} - 2F_{5,1} + F_{6,2} - 6F_{7,3}, \\ 6H_2 &= -6F_{3,0} + 9F_{4,1} + 6F_{5,2} - 3F_{6,3}, \\ 6H_3 &= -3F_{2,0} + 6F_{3,1} - 9F_{4,2} - 6F_{5,3}, \\ 6H_4 &= -6F_{1,0} + F_{2,1} - 2F_{3,2} + 3F_{4,3}. \end{aligned} \tag{12}$$

The sign pattern is checked by reversal: $H_1 \leftrightarrow H_4$, $H_2 \leftrightarrow H_3$, $V_1 \leftrightarrow V_7$, $V_2 \leftrightarrow V_6$, $V_3 \leftrightarrow V_5$, while $V_4 \mapsto -V_4$. Thus magnitudes reverse and exactly the F_4 position changes sign in each pair. Equation (12) puts the tame cyclic coherent-Fano locus inside the persistent locus over $\mathbf{Z}[1/6]$.

We next treat the vertical fibres. In characteristic three the true wild carrier is the cone $\mathrm{Join}(e_2, C_4)$. Any line not through its vertex would project to a line in the quartic normal rational curve, so every contained line is a ruling. Move a ruling to $\langle e_2, e_4 \rangle$. The conditions

$$(a_0, \dots, a_4), (a_1, \dots, a_5) \in \langle e_2, e_4 \rangle$$

give $a_0 = a_1 = a_3 = 0$ and $a_1 = a_2 = a_4 = 0$, leaving only a_5. Undoing the frame gives $f = \mu\nu_5(t)$, including infinity, hence the rank-one/fixed-factor boundary.

In characteristic two the true cyclic component is $\Pi_{\mathrm{cyc}} = (c_0, c_4)$; the other component is supported on $D = 0$. The remaining inseparable degeneration is seen on the *ordered-Hessian chart*: for an ordered pencil basis p_0, p_1, it is the bidegree-$(2, 2)$ equation

$$u^2 \,\mathrm{Hess}^{\mathrm{div}}(p_0) + uv \,\mathrm{PolHess}^{\mathrm{div}}(p_0, p_1) + v^2 \,\mathrm{Hess}^{\mathrm{div}}(p_1) = 0.$$

The tangent quadric has two conic families of generator lines, called the tangent-quadric rulings:

$$\begin{aligned} z_1 = z_4 = 0, \quad z_2 = z_3, \quad z_0z_5 = z_2^2, \\ z_0 = z_5 = 0, \quad z_2 = z_3, \quad z_1z_4 = z_2^2. \end{aligned}$$

The first is the square-quadratic boundary of the persistent Veronese. For the complementary ruling, the two consecutive identities

$$h_{-1}^2 = h_{-2}h_0, \qquad h_1^2 = h_0h_2$$

propagate across their common term and make all five contraction forms proportional, hence rank one. Vertical and horizontal factors are, respectively, collision and fixed-factor components.

It remains to apply the rowspace reduction (3). If the row space lies in the rank-two Hankel component, induct on m: every first contraction has its $(m-1)$-marker row space in the same component, and Proposition 5.14 then puts f in the upper persistent scheme. For a linear modular component $L_S = \langle e_i : i \in S \rangle \subset \Gamma^d E$, the two contractions of an upper basis vector e_q are, up to divided-power normalization, e_q and e_{q-1}. Consequently its coherent lift is the scheme-theoretic kernel

$$\ker\Big(\Gamma^{d+1}E \longrightarrow \mathrm{Hom}(E^\vee, \Gamma^d E/L_S)\Big) = \langle e_q : q, q-1 \in S \rangle.$$

After ℓ lifts its support is therefore exactly the set of q for which $q, q-1, \dots, q-\ell \in S$. For an NRC nucleus, Lucas' theorem computes S from the vanishing binomial digits. Thus the coherent lifts are precisely the Lucas consecutive-support components: the kernel description is scheme-theoretic, commutes with base change, and leaves neither an extra nonlinear component nor an embedded one. The tame projected Veronese contains no line, so its rowspace pullback has rank one. A row space in the characteristic-three cone is a ruling or a point; applying the displayed two-row overlap to each adjacent pair again gives rank one.

Finally, in characteristic two, rowspace containment in Π_{cyc} forces the first and last columns of the catalecticant to vanish:

$$a_0 = \cdots = a_m = 0, \qquad a_4 = \cdots = a_{m+4} = 0.$$

For $m = 1, 2$ these are the known N_3 and $[e_3]$ pullbacks. For $m \geq 3$ the blocks cover every coefficient, so no projective point remains. Hence the characteristic-two modular component descended from the R5 cyclic plane terminates after redundancy seven. This does not erase the fresh higher Lucas carriers of the separate Lucas-carrier analysis; for example $\mathbb{P}\langle e_2, \dots, e_7\rangle \subset \Gamma^9 E$ is nonempty.

The collision and fixed-factor alternatives are already covered by Lemma 5.7 and the gcd/rank boundary. This exhausts the finite bottom component ledger and proves $\mathrm{SC}(j)$ for every $j \geq 6$, together with the asserted termination of the cyclic-plane descendant. $\square$

The rank-one boundary, ordinary rank-two component, collision alternative, and modular pullback are uniform by Proposition 5.14 and Lemmas 5.7 and 5.8. The two inseparable collision possibilities are exactly the characteristic-three R5 case and the characteristic-two R6 case treated in Propositions 4.7 and 6.4. Section 6 independently recovers the redundancies-six and seven base cases.

Theorem 5.11 (One-step polar escape)**.** Let $f \in S_n(\mathbb{F}_q)$ have a one-step lower package as in Definition 5.4. Put

$$\mathcal{H}_\kappa(g, \delta) = 1 + \left\lfloor \left(g + \sqrt{g^2 + \delta - \kappa}\right)^2 \right\rfloor$$

and assume $\delta_\alpha \geq \kappa_\alpha - g_\alpha^2$ for every α. If

$$q \geq \max_\alpha \mathcal{H}_{\kappa_\alpha}(g_\alpha, \delta_\alpha) \quad \text{and} \quad q + 1 > d,$$

then W_f contains a split squarefree member.

Proof. Since $\#\mathbb{P}^1(\mathbb{F}_q) = q + 1 > d$, choose $\lambda \notin Z_f(\mathbb{F}_q)$. On the corresponding lower stratum the recorded point bound gives

$$q + \kappa_\alpha - 2g_\alpha\sqrt{q} > \delta_\alpha,$$

so there is a rational point outside all deletions. It yields a split squarefree lower member avoiding λ. Equation (1) lifts it to a split squarefree member upstairs, so the original system is shallow. $\square$

The theorem isolates the numerical escape and one-step lifting argument. Iterating it requires the contained-component assertion at every intermediate level, not only at the top.

Recursively, the current contraction is chosen outside $\mathcal{P}_j \cup \mathcal{M}_j$; the uniform alternatives and the next SC theorem make its polar line transverse, so a new marker exists off the finite pullback. No inverse-image stability of the declared loci is assumed.

Remark 5.12 (Point-count convention). The correction κ records the exact lower bound available for the chosen model. Every genus-one curve retained in this paper uses the Aubry–Perret bound with $\kappa = 1$; a possible rational singular point is included instead in the deletion degree.

Theorem 5.13 (Uniform transverse threshold)**.** Fix a redundancy $r \geq 6$. If

$$q \geq Q_r := 6r - 15 + \left\lfloor 2\sqrt{6r-17} \right\rfloor,$$

every split-free redundancy-r syndrome belongs to $\mathcal{P}_r \cup \mathcal{M}_r$.

Proof. Iterate the pointed contraction to the R5 cubic pencil. Its off-diagonal curve always has $(g, \kappa) = (1, 1)$, and its singular, diagonal, and branch deletions cost thirteen. Each of the $r - 5$ retained markers removes at most six further ordered pairs. Hence

$$\delta_r = 13 + 6(r-5) = 6r - 17, \qquad \mathcal{H}_1(1, \delta_r) = 6r - 15 + \left\lfloor 2\sqrt{6r-17} \right\rfloor.$$

At a stage of redundancy j with $s = r - j$ old markers, the ordinary pullback costs at most three and each old marker at most three. At the bottom stage $j = 6$, the R5 cyclic/wild pullback costs four by Propositions 6.4 and 6.5, while the collision divisor costs six, giving

$$d_6 \leq 3 + 4 + 6 + 3(r-6) = 3r - 5.$$

For $j \geq 7$, the recursive bad scheme and SC(j) leave only the persistent and modular lower components. Lemmas 5.7 and 5.8 give

$$d_j \leq 3 + 1 + (2j - 6) + 3(r - j) = 3r - j - 2 < 3r - 5.$$

Thus the maximum is attained at the bottom stage, and $Q_r > 3r - 5$ makes the parameter-line inequality $q + 1 > d_j$ automatic. The exact-linear-gcd graph has eight base deletions and at most two per retained marker, hence

$$\delta_r^{\mathrm{lin}} = 8 + 2(r-5) = 2r - 2, \qquad Q_r + 1 - \delta_r^{\mathrm{lin}} = 4r - 12 + \left\lfloor 2\sqrt{6r-17} \right\rfloor > 0.$$

Thus this comparison is weaker for every $r \geq 6$.

Apply Theorem 5.11 successively. If a recursively pointed bad scheme contains the whole polar line, the uniform alternatives above and SC(j) place the syndrome in the declared persistent or modular list. Otherwise a marker remains outside its finite scheme. After reaching the R5 pencil, the displayed point bound supplies a split squarefree cubic avoiding every marker, and Theorem 5.5 lifts it to the original system. ☐

For $r = 5, 6, 7$, the integer cutoffs supplied by the same formula are

$$Q_5 = 22, \qquad Q_6 = 29, \qquad Q_7 = 37;$$

the first prime powers meeting them are 23, 29, 37. Thus the former threshold table consisted of three evaluations of one formula with $Q_r = 6r + O(\sqrt{r})$. The same calculation recovers the verified R6 and R7 parameter bounds 13 and 16. In general the parameter line is nonbinding; the stable-component assertion is supplied uniformly by Theorem 1.3. Propositions 6.7 and Corollary 6.14 discharge the required assertions at redundancies six and seven.

5.2 Contained-component calculations

The ordinary persistent locus admits a uniform contained-line calculation. This does not classify other lower bad schemes, but it closes the persistent part of every level-specific component gate.

Proposition 5.14 (Contained rank-two polar lines)**.** Let $n \geq 5$, let $f \in \mathbb{P}(\Gamma^n E)$, and let

$$A = C_f^{(2)} : \operatorname{Sym}^{n-2} E^\vee \longrightarrow \Gamma^2 E$$

be the three-row catalecticant. Assume the contraction map $E^\vee \to \Gamma^{n-1} E$ is injective, so the first-polar map is defined on all of $\mathbb{P}(E^\vee)$. This is exactly the complement of the rank-one boundary: after a change of coordinates, a nonzero kernel vector is ∂_X, and $\partial_X f = 0$ forces f to be a scalar multiple of $Y^{[n]}$. Such a point lies on the normal rational curve and is already excluded from the split-free locus. Then

$$\Phi_f(\mathbb{P}(E^\vee)) \subset \{b : \operatorname{rank} C_b^{(2)} \leq 2\} \quad \iff \quad \operatorname{rank} A \leq 2$$

scheme-theoretically and in every characteristic. If $\operatorname{rank} A = 3$, the rank-drop parameters form a subscheme of $\mathbb{P}(E^\vee)$ of degree at most three.

Proof. For a nonzero $\lambda \in E^\vee$, multiplication identifies $\operatorname{Sym}^{n-3} E^\vee$ with the hyperplane

$$H_\lambda = \lambda \operatorname{Sym}^{n-3} E^\vee \subset \operatorname{Sym}^{n-2} E^\vee,$$

and functoriality gives

$$C_{\iota_\lambda f}^{(2)} = A|_{H_\lambda}.$$

The forward implication is immediate. Conversely, suppose $\operatorname{rank} A = 3$ and put $K = \ker A$, so $\dim K = n - 4$. If the lower rank is at most two, rank–nullity on the $(n-2)$-dimensional space H_λ gives

$$\dim(K \cap H_\lambda) \geq (n-2) - 2 = n - 4 = \dim K.$$

Thus $K \subset H_\lambda$. Containment for every geometric $[\lambda]$ would imply

$$K \subset \bigcap_{[\lambda] \in \mathbb{P}(E_{\bar{k}}^\vee)} \lambda \operatorname{Sym}^{n-3} E_{\bar{k}}^\vee = 0,$$

because a nonzero binary form cannot be divisible by every linear form. This contradicts $\dim K = n - 4 > 0$.

The maximal minors of $C_{\iota_\lambda f}^{(2)}$ are homogeneous cubics in λ. Vanishing at every geometric point is therefore identical vanishing, proving the scheme-theoretic assertion. In the non-contained case at least one cubic is nonzero, and the common rank-drop scheme lies in its degree-three zero divisor. □

Thus Proposition 5.14 covers the entire non-rank-boundary locus on which the first-polar line is defined; the excluded noninjective locus is the rank-one normal-rational-curve locus just identified. After rank one and rational split secants are removed as shallow, the upper rank-two locus consists of the persistent tangent and conjugate-secant families. On a persistent fiber the nonsplit torus acts by $z \mapsto z^n$, giving T/T^n modulo inversion and Frobenius, while the tangent unipotent cocycle is $z \mapsto z + nu$.

Modular loci are equally explicit. If $\mathcal{N}$ is a nucleus in the lower divided-power module, the complete space whose first-polar line lies in $\mathcal{N}$ is

$$\mathcal{M}_n(\mathcal{N}) = \mathbb{P} \ker \left(\Gamma^n E \longrightarrow \operatorname{Hom}(E^\vee, \Gamma^{n-1} E / \widetilde{\mathcal{N}}) \right).$$

Lucas' theorem computes the nucleus coordinates, and this displayed kernel computes every coherent lift without an ambient syndrome census. At redundancy r, if $p > r - 1$, Lucas' theorem leaves no nucleus coordinate, so $\mathcal{M}_r$ is empty. This removes the modular term only; the R5 cyclic carrier survives in large characteristic.

Remark 5.15 (Scope of the contained calculations). The modular contraction kernel is an exact finite linear-algebra construction. Proposition 5.14 supplies the ordinary persistent component uniformly, and Lemma 5.7 supplies the collision alternative. The remaining arbitrary-level question is SC(j): whether another contained component occurs.

6 Redundancies six and seven: complete and gated results

6.1 Redundancy six

For

$$f = (a_0 : \cdots : a_5) \in \mathbb{P}(\Gamma^5 E), \qquad W_f = \ker \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_1 & a_2 & a_3 & a_4 & a_5 \end{pmatrix},$$

the system W_f is a net of quartics off the quintic normal rational curve. It is split-free exactly when it has no completely split squarefree quartic.

Proposition 6.1 (Persistent net and orbit law)**.** If $\gcd(W_f) = Q$ has degree two, then

$$W_f = Q \operatorname{Sym}^2 E^\vee.$$

The split-free persistent locus has size $q(q+1)^2/2$. On the tangent fiber the unipotent coordinate is $x \mapsto x + 5u$; on the irreducible quadratic fiber the norm-one torus acts through $z \mapsto z^5$, with inversion and coefficient Frobenius acting on T/T^5.

Proof. For fixed Q, the syndromes annihilating $Q \operatorname{Sym}^3 E^\vee$ form a projective line. If Q is irreducible, all $q+1$ points remain; if $Q = \lambda^2$, one endpoint has dependent Hankel rows, leaving q points. Hence

$$q(q+1) + \frac{q(q-1)}{2}(q+1) = \frac{q(q+1)^2}{2}.$$

Normalize a square to U^2. The nondegenerate fiber is $[e_4 + xe_5]$, and direct degree-five substitution gives $x \mapsto x + 5u$. For an irreducible Q, its stabilizer identifies the $q+1$-point fiber with $T = \{z \in \mathbb{F}_{q^2}^\times : z^{q+1} = 1\}$ and acts by fifth powers; the normalizer inverts z, while coefficient Frobenius multiplies T/T^5 by the characteristic. This proves the stated orbit law and the characteristic-five tangent splitting. ☐

Lemma 6.2 (Exact linear gcd is shallow)**.** Let $q \geq 7$. If $\gcd(W_f)$ has degree one, then W_f contains a split squarefree quartic.

Proof. Write $W_f = LV$, where L is rational and $V \subset \operatorname{Sym}^3 E^\vee$ has vector dimension three. It suffices to find in V a product of three distinct rational factors, none equal to L. Move the root of L to infinity and let ℓ span $V^\perp$. On the cubic with distinct finite roots x, y, z, write

$$\ell\big((T - xU)(T - yU)(T - zU)\big) = c_0 + c_1(x + y + z) + c_2(xy + xz + yz) + c_3 xyz =: P(x, y, z).$$

Suppose that this value is nonzero for every distinct triple. For fixed distinct x, y, write $P = A(x, y) + zB(x, y)$, where

$$B(x, y) = c_1 + c_2(x + y) + c_3 xy.$$

If $B(x, y) \neq 0$, the unique zero in z must be x or y. Consequently

$$B(x, y)P(x, y, x)P(x, y, y) = 0$$

for every $x \neq y$; it also vanishes when $B(x, y) = 0$. This polynomial has degree at most four in each variable. Since $q - 1 > 4$, first varying $y \neq x$ and then x shows that it is the zero polynomial. The polynomial ring is a domain. Neither $P(x, y, x)$ nor $P(x, y, y)$ vanishes identically unless $\ell = 0$, as comparison of the coefficients of $1, x, y, x^2, xy, x^2y$ shows. Hence $B = 0$, so $c_1 = c_2 = c_3 = 0$.

Thus V is the hyperplane of cubics vanishing at infinity, and every member of V is divisible by L. This would make $\gcd(W_f)$ divisible by L^2, contrary to its exact degree. Therefore a required cubic exists, and multiplying it by L gives a split squarefree member of W_f. ☐

Proposition 6.3 (First-polar line and secant degree)**.** For $r \in \mathbb{F}_q$ put

$$b(r) = (a_1 - ra_0, \ldots, a_5 - ra_4), \qquad b(\infty) = (a_0, \ldots, a_4).$$

Then

$$(T - rU)g \in W_f \iff g \in W_{b(r)}.$$

If W_f has trivial gcd, its polar line meets the lower secant cubic in degree at most three.

Proof. The first equivalence follows by expanding the two Hankel equations of $(T - rU)g$. Let

$$C_f = \begin{pmatrix} a_0 & a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 & a_4 \\ a_2 & a_3 & a_4 & a_5 \end{pmatrix}.$$

The lower catalecticant equals

$$C_f \begin{pmatrix} -r & 0 & 0 \\ 1 & -r & 0 \\ 0 & 1 & -r \\ 0 & 0 & 1 \end{pmatrix}.$$

Cauchy–Binet gives the secant determinant

$$D_f(r) = -\Delta_{012}r^3 + \Delta_{013}r^2 - \Delta_{023}r + \Delta_{123}.$$

It vanishes identically exactly when $\operatorname{rank} C_f \leq 2$, equivalently when W_f has quadratic gcd. In the trivial-gcd case it therefore has at most three zeros. ☐

Proposition 6.4 (Cyclic and collision degrees)**.** For a trivial-gcd net:

(i) in characteristic different from two and three, the lower tame cyclic carrier is a binomially rescaled Veronese surface of degree four and contains no polar line;

(ii) the pointed ramification condition has degree at most six and does not vanish identically;

(iii) an S_3 lower pencil with the marked root deleted has total deletion degree at most nineteen.

Proof. In syndrome coordinates the tame cyclic surface is

$$\left\{ \operatorname{diag}(1,4,6,4,1)^{-1}[Q^2] : [Q] \in \mathbb{P}(\operatorname{Sym}^2 E^\vee) \right\}.$$

When 2 and 3 are invertible this is the Veronese surface, of degree four, and contains no line. Characteristic three has a different wild closure, treated in Proposition 6.5. For (ii), Lemma 5.7 with $j = 6$ gives the degree-six bound except possibly in characteristic two. In that characteristic, inseparability would identify W_f, after a change of coordinates, with the full pure-square space $\langle T^4, T^2U^2, U^4 \rangle$. Its annihilator is $\langle e_1^\vee, e_3^\vee \rangle$. For the two consecutive Hankel rows this would require simultaneously

$$a_0 = a_2 = a_4 = 0, \qquad a_1 = a_3 = a_5 = 0,$$

contrary to $f \neq 0$. Thus the bound holds in every characteristic.

Finally the R5 off-diagonal curve has deletion degree thirteen, including its possible rational singular point. The unique cubic through a specified marker contributes at most six ordered pairs, giving $(g, \delta, \kappa) = (1, 19, 1)$ and

$$q + 1 - 2\sqrt{q} > 19.$$

Its first prime-power solution is 29. ☐

Proposition 6.5 (Modular contained components)**.** In characteristic three the lower wild carrier is the degree-four cone

$$\mathrm{Join}(e_2, C_4),$$

and no polar line of a trivial-gcd net is contained in it. In characteristic two the cyclic carrier is the plane

$$\Pi_{\mathrm{cyc}} = \mathbb{P}\langle e_1, e_2, e_3\rangle,$$

whose unique contained polar-line component is

$$\mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle.$$

Proof. Every line on the wild cone is a ruling. By equivariance it suffices to compare the two consecutive rows on one ruling; in the frame $\langle e_2, e_4\rangle$ the conditions

$$(a_0, \ldots, a_4),\ (a_1, \ldots, a_5) \in \langle e_2, e_4\rangle$$

force the overlapping coordinates successively to vanish. Undoing the frame gives $f = \mu\nu(t)$, including $f = \mu e_5$ at infinity. These are fixed-factor/rank-boundary syndromes, so no trivial-gcd polar line is contained in the wild cone.

In characteristic two, direct substitution gives

$$g \cdot e_2 = [0 : \delta\gamma : \alpha\delta + \beta\gamma : \alpha\beta : 0],$$

whose closure is Π_{cyc}. The consecutive-row overlap is now literal: the first row lies in the plane exactly when $a_0 = a_4 = 0$, and the second exactly when $a_1 = a_5 = 0$. Hence a polar line lies in the plane exactly when

$$a_0 = a_1 = a_4 = a_5 = 0,$$

namely when $f \in \mathbb{P}\langle e_2, e_3\rangle$. This proves both existence and uniqueness of the contained binary component. □

Proposition 6.6 (Binary nucleus arithmetic)**.** The scheme $\mathcal{N}_3(\mathbb{F}_q)$ is one projective-semilinear orbit of $q + 1$ points. It is split-free over $q = 2^m$ exactly when m is odd.

Proof. For the representative e_3,

$$W_{e_3} = \langle 1, t, t^4\rangle.$$

If $C = 0$, the member $A+Bt+Ct^4$ has degree at most one and cannot have four distinct roots. Assume $C \neq 0$. The root differences of $A+Bt+Ct^4$ then lie in the kernel of $u \mapsto u^4+(B/C)u$. Four distinct rational roots exist exactly when all roots of $u^3 = B/C$ are rational, equivalently when $3 \mid q - 1$, which for $q = 2^m$ means m is even. The stabilizer of e_3 is the Borel of order $q(q-1)$, so the orbit has $q+1$ points; its lower-unipotent transforms and endpoint are precisely $\mathcal{N}_3(\mathbb{F}_q)$. □

Proposition 6.7 (Exhaustive R6 contained-component classification)**.** For a redundancy-six syndrome, the alternatives in Definition 5.9 are exhausted as follows:

(i) a noninjective contraction map is the rank-one normal-rational- curve boundary and is shallow;

(ii) a polar line contained in the lower secant cubic comes from the upper rank-two catalecticant locus; after its shallow rational secants are removed, this is the persistent tangent and conjugate-secant locus;

(iii) no trivial-gcd polar line is contained in the tame cyclic surface or the characteristic-three wild cone;

(iv) in characteristic two, containment in the cyclic plane occurs exactly for $f \in \mathcal{N}_3 = \mathbb{P}\langle e_2, e_3\rangle$;

(v) on the remaining trivial-gcd locus the collision divisor is not identically zero.

Consequently the complete nonshallow contained list is the persistent locus together with $\mathcal{N}_3$ in characteristic two. There is no unlisted contained component.

Proof. The noninjective and secant-cubic alternatives are Proposition 5.14 and Proposition 6.3. The tame cyclic and collision assertions are Proposition 6.4; the wild and binary specializations are Proposition 6.5. These are all the clauses of Definition 5.9: contraction indeterminacy, containment in each component of the lower bad scheme, and identical collision. Hence the list is exhaustive. □

Proposition 6.8 (R6 one-step exhaustion)**.** Assume $q \geq 29$. Then every split-free redundancy-six syndrome lies in the persistent quadratic-gcd locus or, when $q = 2^m$ with odd m, in $\mathcal{N}_3(\mathbb{F}_q)$.

Proof. The complete contained branch is Proposition 6.7. The common-factor trichotomy is exhaustive. A quadratic gcd gives the persistent tangent and conjugate-secant loci, together with the shallow rational secant; an exact linear gcd is shallow by Lemma 6.2. It remains to treat a trivial-gcd net.

Its first-polar line is not contained in the lower secant cubic, by Proposition 6.3. The pullback of that cubic has degree at most three. The tame cyclic surface has degree four and contains no polar line. In characteristic three its wild replacement has the same degree and no contained trivial-gcd polar line; in characteristic two the replacement is a plane, whose only contained component is $\mathcal{N}_3$, by Proposition 6.5. Finally, Proposition 6.4 gives a nonzero collision divisor of degree at most six. Outside the contained binary component, the excluded parameter scheme therefore has degree

$$d \leq 3 + 4 + 6 = 13.$$

For every remaining marker, the lower cubic pencil has exact linear gcd or geometric monodromy S_3. In the first case the graph argument of Proposition 4.3 is pointed as follows: besides its eight original deletions, remove the at most two graph points whose split quadratic contains the new marker. Since $q + 1 > 10$, a split squarefree cubic avoiding that marker remains. Here the fixed gcd λ_0 differs from the marker r. Indeed, $\lambda_0 = r$ says that every cubic in $W_{b(r)}$ vanishes at r, so every lifted quartic has a double root there; by Lemma 5.6, this is exactly a point of the degree-six collision divisor already placed in Z_f, so no additional deletion budget is needed. In the S_3 case, Lemma 4.6 supplies the geometrically integral off-diagonal curve. Its singular, diagonal, and branch deletions have degree at most thirteen, and deleting the unique cubic through the marker removes at most six ordered pairs. Thus

$$(g, \delta, \kappa) = (1, 19, 1), \qquad q + 1 - 2\sqrt{q} > 19$$

for every $q \geq 29$. Explicitly,

$$\mathcal{H}_1(1, 19) = 1 + \left\lfloor (1 + \sqrt{19})^2 \right\rfloor = 29, \qquad 30 - 2\sqrt{29} > 19.$$

These data form the one-step package of Definition 5.4, with $d = 13$, so Theorem 5.11 produces a split squarefree quartic.

This is one field-order threshold, not three characteristic-dependent estimates: the first characteristic-three and characteristic-two prime powers above it happen to be 81 and 32, respectively. On $\mathcal{N}_3$, Proposition 6.6 gives the asserted odd-degree split-free orbit and proves shallowness in even extension degree. $\square$

The covering-radius gate is complete: Seroussi–Roth gives $\rho(\mathrm{PRS}(q-5)) = 5$ for $q \geq 9$, while Certificate R6 checks it directly at $q = 7, 8$.

Theorem 6.9 (Redundancy six)**.** For every prime power $q \geq 7$, the deep syndromes of $\mathrm{PRS}(q-5)$ are the persistent locus of size $q(q+1)^2/2$, together with:

(i) respectively $18, 11, 4, 2, 1$ exceptional $\mathrm{PGL}_2(q)$ orbits at $q = 7, 8, 9, 11, 13$;

(ii) one characteristic-two nucleus orbit of size $q+1$ when $q = 2^m$ and odd $m \geq 5$.

The lower bound $m \geq 5$ avoids double counting: the remaining odd case $m = 3$, namely $q = 8$, is already the $|\mathcal{O}| = 9$ row of the exceptional table. There are no other exceptional orbits. The complete persistent orbit law is T/T^5 modulo inversion and Frobenius, with a tangent fixed/nonzero split exactly in characteristic five. At $q = 7, 8, 9, 11, 13$, the exceptional direction totals are

$$5152, \quad 4713, \quad 1800, \quad 792, \quad 546,$$

and the corresponding total numbers of deep syndrome directions are

$$5376, \quad 5037, \quad 2250, \quad 1584, \quad 1820.$$

Proof. Propositions 6.7 and 6.8 give the complete conceptual exhaustion for every prime power $q \geq 29$. Certificate R6 exhausts

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27\},$$

finds exceptional projective-orbit counts $18, 11, 4, 2, 1$ only at the five displayed fields, and no further orbit at the remaining fields. Together with the separate radius gate, this covers every prime power $q \geq 7$. $\square$

The listed sources are disjoint. The persistent locus has quadratic gcd, every finite exceptional row has trivial gcd, and the separate nucleus term occurs only for binary fields beyond the finite table.

The five small tables are exhaustive computations. For an exceptional net W_f, let $I_f(r)$ be the number of irreducible cubic members of the quotient pencil $W_{b(r)}$, including the infinity chart, and define its shared-root pair count by

$$E_f = \sum_{r \in \mathbb{P}^1(\mathbb{F}_q)} \binom{I_f(r)}{2}.$$

In odd characteristic let

$$F_f(T, U) = \sum_{i=0}^{5} a_i T^{5-i} U^i$$

be the binary quintic attached to $f = (a_0 : \cdots : a_5)$, and let $\tau_5(f)$ be its rational factorization type; for example, $1^3 + 2$ denotes a triple rational factor and an irreducible quadratic factor. Their semilinear orbit counts are $18, 5, 2, 2, 1$; E_f, together in odd characteristic with $\tau_5(f)$, separates the classes up to precisely Frobenius fusion. Table 6 prints the canonical representatives, stabilizers, orbit sizes, and fusion lengths.

Table 6: Exceptional redundancy-six classes. Each row is one projective-semilinear class. The columns give a canonical syndrome representative, the size and stabilizer order of each constituent $\mathrm{PGL}_2(q)$-orbit, the number ϕ of such orbits fused by coefficient Frobenius, the shared-root pair count E, and the binary-quintic factor type τ_5 in odd characteristic.

q	representative	$\lvert\mathcal{O}\rvert$	$\lvert G_f\rvert$	ϕ	E	τ_5
7	$[0:0:0:1:0:1]$	168	2	1	7	1^3+2
7	$[0:0:1:0:3:1]$	336	1	1	19	1^2+1+2
7	$[0:0:1:0:3:2]$	336	1	1	15	1^2+3
7	$[0:1:0:0:0:1]$	168	2	1	10	$1+2+2$
7	$[0:1:0:0:2:0]$	112	3	1	33	$1+1+3$
7	$[0:1:0:1:1:2]$	336	1	1	9	$1+4$
7	$[0:1:0:1:1:5]$	336	1	1	9	$1+1+3$
7	$[0:1:0:1:3:0]$	336	1	1	21	$1+1+1+2$
7	$[0:1:0:1:3:3]$	336	1	1	7	$1+1+3$
7	$[0:1:0:1:3:6]$	336	1	1	14	$1+1+3$
7	$[0:1:0:3:0:5]$	168	2	1	11	$1+4$
7	$[0:1:0:3:0:6]$	168	2	1	20	$1+4$
7	$[0:1:0:3:2:1]$	336	1	1	13	$1+1+3$
7	$[1:0:0:0:1:2]$	336	1	1	7	$1+4$
7	$[1:0:0:0:1:3]$	336	1	1	23	5
7	$[1:0:0:3:1:3]$	336	1	1	13	$1+4$
7	$[1:0:1:0:5:2]$	336	1	1	8	$1+4$
7	$[1:0:1:0:6:2]$	336	1	1	8	5
8	$[0:0:0:1:0:0]$	9	56	1	0	–
8	$[0:1:1:0:2:1]$	504	1	3	20	–
8	$[0:1:1:0:2:5]$	504	1	3	19	–
8	$[1:0:0:1:3:6]$	504	1	3	14	–
8	$[1:0:1:3:5:2]$	168	3	1	54	–
9	$[0:1:0:0:0:4]$	180	4	2	28	$1+4$
9	$[0:1:0:1:4:2]$	720	1	2	29	$1+4$
11	$[0:0:0:1:0:0]$	132	10	1	60	1^3+1^2
11	$[0:1:0:2:0:10]$	660	2	1	30	$1+4$
13	$[0:1:0:0:0:2]$	546	4	1	60	$1+4$

For $q=8$, integers are binary polynomial-basis codes with $\alpha^3=\alpha+1$; for $q=9$, they are ternary polynomial-basis codes with $\alpha^2=-1$. Thus $\sum\phi=(18,11,4,2,1)$ and $\sum\phi\lvert\mathcal{O}\rvert=(5152,4713,1800,792,546)$, in the field order $7,8,9,11,13$.

6.2 Redundancy seven

For

$$f=(a_0:\ldots:a_6)\in\mathbb{P}(\Gamma^6E),\qquad W_f=\ker\begin{pmatrix}a_0&a_1&a_2&a_3&a_4&a_5\\a_1&a_2&a_3&a_4&a_5&a_6\end{pmatrix},$$

the system W_f is a four-dimensional web of quintics. For an affine marker r the contractions are

$$b(r)=(a_1-ra_0,\ldots,a_6-ra_5).$$

Coefficient collection gives

$$(t-r)g \in W_f \quad \Longleftrightarrow \quad g \in W_{b(r)};$$

the lift is squarefree exactly when $g(r) \neq 0$.

The contracted net $W_{b(r)}$ has quadratic gcd exactly when its three-row catalecticant has rank at most two. Unless the first-polar line is contained in that rank-two locus, these parameters form the degree-at-most-three intersection of Proposition 6.3; the contained case is Corollary 6.14.

Proposition 6.10 (Complete two-marker lower package)**.** Let W_h be a trivial-gcd quartic net and let x be a prescribed forbidden root. Assume in characteristic two that $h \notin \mathcal{N}_3$. For every prime power $q \geq 37$, the net contains a split squarefree quartic avoiding x.

Proof. By equivariance choose coordinates in which x is finite; all parameter equations below are then homogenized in s, so the infinity chart is included. Choose the second contraction marker s. The unavailable values of s comprise: $s = x$, of degree one; at most three intersections with the lower secant cubic, by Proposition 6.3; at most four intersections with the cyclic surface or its characteristic-three wild replacement; and at most six points of the collision divisor, by Propositions 6.4 and 6.5. In characteristic two the cyclic replacement is a plane, so its pullback has degree at most one unless the polar line is contained in it, the excluded case $h \in \mathcal{N}_3$, by Proposition 6.5. We must also exclude the parameters for which the lower pencil has fixed factor x. They are cut out by the 3-minors obtained by appending the fixed evaluation row $(1, x, x^2, x^3)$ to the two Hankel rows of $b(s)$, and hence have degree at most two in s. These minors are not all zero. If they vanished identically, then

$$W_h \cap \{g : g(s) = 0\} \subseteq \{g : g(x) = 0\}$$

for every geometric s. Given $g \in W_h$, choose a geometric root s of g; then $g(x) = 0$. Thus x would divide every member of W_h, contradicting the trivial-gcd hypothesis. Hence the complete second-step parameter scheme has degree at most

$$1 + 3 + 4 + 6 + 2 = 16 < q + 1.$$

Choose s outside it.

The resulting cubic pencil is neither persistent nor cyclic. If it has exact linear gcd λ_0, then $\lambda_0 \notin \{x, s\}$: equality with x is excluded by the fixed-factor minors above, while equality with s is the pointed collision excluded by Proposition 6.4. The graph proof of Proposition 4.3 has eight original deletions; avoiding the two additional markers costs at most two graph points each. Since $q + 1 > 12$, a split squarefree cubic avoiding x and s remains.

Otherwise the bottom cover has geometric monodromy S_3. Its off-diagonal curve is geometrically integral of arithmetic genus one. The singular, diagonal, and branch deletions cost at most thirteen, and the fibers through x and s cost at most six ordered pairs each. Hence

$$(g, \delta, \kappa) = (1, 25, 1), \qquad q + 1 - 2\sqrt{q} > 25,$$

whose first prime-power solution is 37. In either case the cubic lifts through s to a split squarefree quartic in W_h avoiding x. □

Proposition 6.11 (Exact-gcd-one avoidance)**.** Suppose the lower net is $\ell_x U$, where $U = \ker \lambda \subset \operatorname{Sym}^3 E^\vee$ has dimension three. If $x = r$, the parameter belongs to the collision divisor. If $x \neq r$ and $q \geq 16$, then U contains a split cubic avoiding both x and r.

Proof. If $x = r$, every lifted quintic has the repeated factor $(t-r)^2$. Lemma 5.6 identifies this with the degree-eight collision divisor of Proposition 6.12. This is the unavailable branch, so assume $x \neq r$. Put $S = \mathbb{P}^1(\mathbb{F}_q) \setminus \{x, r\}$. For ordered distinct $u, v \in S$, the map $w \mapsto \lambda(\ell_u \ell_v \ell_w)$ is linear. If it vanishes identically, choose any remaining w; otherwise it has one rational zero. The equations $w = x, r, u, v$ have respective bidegrees $(1,1), (1,1), (2,1), (1,2)$ in (u, v). None is identically zero: the first two would add a common factor, and the last two are excluded because double-root cubics span $\mathrm{Sym}^3 E^\vee$. A nonzero bidegree-(a,b) divisor on $\mathbb{P}^1 \times \mathbb{P}^1$ has at most $(a+b)(q+1)$ rational points. Thus the bad pairs number at most $10(q+1)$, whereas $(q-1)(q-2) > 10(q+1)$ for $q \geq 16$. □

Proposition 6.12 (Collision divisor)**.** After fixed factors are removed, the collision divisor of the moving g^3_5 has degree at most eight.

Proof. This is Lemma 5.7 with $j = 7$: the series is separable and $2j - 6 = 8$. □

The characteristic-two nucleus line at redundancy six has exactly one coherent lift, the central point e_3, with

$$W_{e_3} = \langle 1, t, t^4, t^5 \rangle.$$

It is deep for odd m; for even m, the homogenization of $t^4 + t$ has the five roots $\mathbb{P}^1(\mathbb{F}_4)$.

Proposition 6.13 (Central binary lift)**.** The complete inverse image of the R6 nucleus line under consecutive sextic contraction is $[e_3]$. It is split-free over $\mathbb{F}_{2^m}$ exactly when m is odd.

Proof. Putting both consecutive rows in $\mathbb{P}\langle e_2, e_3 \rangle$ gives $a_0 = a_1 = a_2 = a_4 = a_5 = a_6 = 0$. Thus only e_3 remains, with $W_{e_3} = \langle 1, t, t^4, t^5 \rangle$. For odd m, contraction at a root of a hypothetical split quintic would give a split quartic on the R6 nucleus line, contradicting Proposition 6.6. For even m, $\mathbb{F}_4 \subset \mathbb{F}_q$ and the homogenization of $t^4 + t$ has its four affine $\mathbb{F}_4$ roots and the simple root at infinity. □

Corollary 6.14 (Exhaustive R7 contained-component classification)**.** The assertion $\mathrm{CC}(6,1)$ is exhaustive in every characteristic. A noninjective contraction is the shallow rank-one boundary. Containment in the lower persistent locus gives upper catalecticant rank at most two; after the shallow rational secants are removed, these are exactly the persistent tangent and conjugate-secant families. The complete lift of the only additional lower component, the characteristic-two nucleus line, is the central point e_3. The remaining collision divisor is nonzero. Thus the nonshallow contained list is precisely the persistent locus and, in characteristic two, e_3; there is no further contained component.

Proof. Apply Proposition 5.14 with $n = 6$. The three rational types of the quadratic recurrence are split squarefree, repeated rational, and irreducible. They give respectively the shallow rational secant, tangent, and conjugate-secant loci. The remaining lower binary nucleus line has the separate complete lift computed in Proposition 6.13. Proposition 6.12 shows that the collision divisor is nonzero. These are respectively the contraction-indeterminacy, lower-component-containment, and identical-collision alternatives of Definition 5.9, so the list is exhaustive. □

Theorem 6.15 (Redundancy-seven classification)**.** Certificate R7 unconditionally classifies every prime power $q \leq 32$ in its displayed domain. For every prime power $q \geq 13$, the split-free syndromes at redundancy seven are the persistent locus of size $q(q+1)^2/2$, together with the central nucleus point when $q = 2^m$ and m is odd. At $q = 7, 8, 9, 11$ the additional split-free orbit-size profiles, after removing the persistent locus and also the central nucleus singleton at $q = 8$, are:

$$\begin{array}{r|l} 7 & 56,\ 84^5,\ 112^2,\ 168^{45},\ 336^{141} \\ 8 & 63,\ 72,\ 84^3,\ 168^4,\ 252^{24},\ 504^{86} \\ 9 & 180^3,\ 240^6,\ 360^{18},\ 720^{27} \\ 11 & 264^2,\ 440,\ 660^2. \end{array}$$

This is the complete all-field split-free classification. Since Seroussi–Roth gives $\rho(\mathrm{PRS}(q-6)) = 6$ for every prime power $q \geq 11$, it is also the complete deep-hole classification in that range. At $q = 7, 8, 9$ the table is not promoted to a code deep-hole classification without a separate covering-radius calculation. The persistent orbit law is T/T^6 modulo inversion and Frobenius, with tangent splitting in characteristics two and three.

Proof. For fixed irreducible quadratic Q, the persistent fiber has $q+1$ points, while for $Q = \lambda^2$ it has q non-rank-one points. There are $q(q-1)/2$ irreducible quadratics and $q+1$ rational squares. Hence the persistent count is

$$\frac{q(q-1)}{2}(q+1) + (q+1)q = \frac{q(q+1)^2}{2}.$$

For $q \geq 37$, choose a first marker outside the at most three catalecticant intersections, eight collision parameters, and, in characteristic two, one intersection with the nucleus line. If the contracted quartic net has trivial gcd, Proposition 6.10 supplies the complete second one-step package: it excludes the cyclic/wild carrier and fixed-gcd/marker coincidences before applying either the pointed linear-gcd graph or the S_3 cover. If the contracted net has exact linear gcd, Proposition 6.11 applies; its excluded $x = r$ case is already among the eight collision parameters. Corollary 6.14 leaves exactly the persistent locus and Proposition 6.13. Certificate R7 covers

$$q \in \{7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32\}$$

by an independent five-secant replay. The radius theorem promotes the split-free list only from $q \geq 11$. □

For the four finite rows Certificate R7 contains the canonical representative, stabilizer, persistent/central flags, and Frobenius link of every orbit. The displayed size profile is only a compact summary; repeated sizes are distinguished by those canonical records. There are respectively $194, 119, 54, 5$ exceptional representatives at $q = 7, 8, 9, 11$; the complete lists are printed in `supplement/CLASSIFICATION-RECORDS.md`, while the matching JSON is the machine-readable record. The following gives one printed canonical representative for every orbit-size block in the displayed profile; repeated orbits within a block are distinguished in the full list:

q	$\lvert\mathcal{O}\rvert$	representative
7	56	$[0:0:0:0:1:0:0]$
	84	$[0:1:0:0:0:1:0]$
	112	$[0:0:1:0:0:2:0]$
	168	$[0:0:0:0:1:0:1]$
	336	$[0:0:0:1:0:1:1]$
8	63	$[0:0:0:1:0:0:1]$
	72	$[0:0:0:0:1:0:0]$
	84	$[0:1:1:1:1:3:6]$
	168	$[1:0:1:1:3:7:7]$
	252	$[0:0:1:0:1:0:0]$
	504	$[0:0:1:0:1:0:1]$
9	180	$[0:1:0:0:0:4:0]$
	240	$[0:0:1:0:1:1:1]$
	360	$[0:1:0:1:3:3:2]$
	720	$[0:0:1:1:0:2:2]$
11	264	$[0:1:0:0:0:0:2]$
	440	$[1:0:1:1:4:3:3]$
	660	$[1:0:1:0:1:4:8]$.

For $q = 8, 9$, the coordinates use the same polynomial-basis encoding specified below Table 6.

The $q = 19$ lower net

$$W = \langle 1, t^3, t^4 \rangle$$

shows why the parameter must remain marked. Its split squarefree members are exactly

$$U(T^3 - uU^3), \qquad u \in (\mathbb{F}_{19}^{\times})^3 = \{1, 7, 8, 11, 12, 18\}.$$

All six contain the marked point at infinity. The obstruction is pointed rather than intrinsic to W: at the marker 0, for example, $U(T^3 - U^3)$ is an admissible member. No coherent sextic polar line lies in the infinity-pointed orbit. Thus the R7 proof escapes over the marker parameter; classifying unpointed bad lower fibers would retain a false exception.

Table 7: Field-range closure ledger. "Certificate" means a complete finite classification with canonical representatives, not merely an orbit count.

Result	prime powers	closure mechanism
R5	$7, 8, 9, 11, 13, 16, 17, 19$	direct census and independent replay
	$q \geq 23$	cubic-cover geometry, Lemma 4.6
R6	$7, 8, 9, 11, 13, 16$	direct syndrome/radius census
	$17, 19, 23, 25, 27$	finite structural bridge certificate
	$q \geq 29$	marked-cover point bound and contained components
R7	$7, 8, 9, 11$	direct split-free census
	$13, 16, 17, 19, 23, 25, 27, 29, 31, 32$	coherent-polar finite bridge certificate
	$q \geq 37$	marked-cover point bound and contained components

The R7 rows classify split-free syndromes. Conversion to code deep-hole directions uses the separate covering-radius gate and is made only for $q \geq 11$. The public certificate schema and exact replay boundaries are given in the supplement; the table does not promote orbit-size profiles to identifiers.

There is no prime power in any of the open numerical intervals $(19, 23)$, $(27, 29)$, or $(32, 37)$. Thus adjacent finite and geometric rows in Table 7 leave no unlisted field order.

7 Verification and formal boundary

No geometric classification theorem in this paper is claimed to be formalized in Lean. The formal layer checks coordinate identities and conditional synthesis implications with the component geometry, rational-point existence, group actions, and certificate semantics kept as explicit inputs.

The conceptual theorems above are accompanied by deterministic evidence bundles. Table 8 records which claims use exhaustive computation. Each public label resolves, through the supplement schema, to its generator, certificate, replay where stated, and SHA-256 manifest. Table 1 gives the corresponding claim-level mathematical dependencies.

Table 8: Public verification map. Exact filenames, commands, hashes, domains, and stop conditions belong to the electronic supplement.

Public label	Verified domain	Replay and trust boundary
Certificate R5	nineteen finite fields	independent pointwise and orbitwise replay; canonical representatives and exhaustion records
Certificate R6	direct scan through $q = 16$; finite bridge below 29	independent syndrome replay and separate radius gate
Certificate R6-NF	small exceptional normal forms	deterministic normal-form reconstruction; not a second implementation
Certificate R7	$q = 7, 8, 9, 11$ and finite bridge below 37	independent five-secant/orbit checks; split-free and radius flags remain separate
Certificate SC	integral stable-component model in every characteristic	exact polynomial identities and finite-field factorization replay; Singular checks saturation and vertical primary decompositions

Table 6 is a deterministic projection of the hash-pinned public R6 and R6-NF records. The top-level verifier regenerates it and checks byte-for-byte equality with the included source.

The public labels are defined by `supplement/CERTIFICATE-SCHEMA.md`. The paper-local bundle, literal replay commands, toolchain locks, hashes, byte counts, and working directories are recorded in `supplement/REPRODUCING.md`, `supplement/EVIDENCE-MANIFEST.json`, and `supplement/RELEASE-MANIFEST.md`. The single entry point `python3 supplement/verify.py` checks the bundle manifest, classification records and their expected hashes, and the local PDF and supplement-artifact rows printed in the release manifest; its `--replay` option runs the paper-local Python replays.

The paper-facing Lean gate checks coordinate algebra and conditional synthesis, not the geometric classifications or rational-point arguments. The exact declarations, hypotheses, axiom audit, and reproduction command are listed in `supplement/LEAN-STATEMENTS.md`.

Data, code, and disclosure

The electronic supplement accompanying this manuscript contains the source, canonical classification representatives and stabilizers, certificate schemas, generators, independent replays where available, expected hashes, toolchain locks, and a top-level verification command. Section 7 assigns each computational or formal claim its exact trust route.

8 Scope and open problems

The contained/transverse fork gives a uniform all-level theorem. Theorem 1.3 supplies the scheme-theoretic component classification without a field-size hypothesis, and Theorem 5.13 supplies the arithmetic threshold.

Corollary 8.1 (Large-characteristic stable list)**.** Let $r \geq 6$ and $q \geq Q_r$, and suppose $\mathcal{M}_r$ is empty. Then the split-free directions are exactly the persistent tangent and conjugate-secant families. If the covering radius is $r-1$, this is also the projective deep-hole list. The modular locus is empty when $\operatorname{char}\mathbb{F}_q > r-1$, so the conclusion applies in particular over every prime field with $q \geq Q_r$.

Proof. Theorem 5.13 leaves $\mathcal{P}_r \cup \mathcal{M}_r$, and Lucas' theorem makes $\mathcal{M}_r$ empty. The persistent families are split-free. □

Proposition 6.7 and Corollary 6.14 prove the first two assertions unconditionally. The characteristic hypothesis removes the modular term, not the large-characteristic R5 cyclic carrier; the theorem eliminates that carrier on the transverse branch.

The result is an arbitrary-redundancy containment theorem, not by itself a complete deep-hole classification.

1. Redundancy five is complete for every $q \geq 7$.

2. Redundancy six is a complete all-field deep-hole classification. Redundancy seven has a complete all-field split-free classification; its $q = 7, 8, 9$ covering-radius gate remains separate.

3. At every redundancy $r \geq 6$ and $q \geq Q_r$, every split-free direction lies in the persistent or modular locus. The binary cyclic-plane descendant terminates after redundancy seven, but fresh higher Lucas carriers remain possible. Fields below Q_r and the covering-radius premise remain separate.

In particular, the paper does not settle the general Reed–Solomon deep-hole conjecture: it neither proves the covering-radius premise nor classifies the fields below the transverse threshold.

The last sporadic fields at redundancies five, six, and seven are $19, 13, 11$, while the corresponding point-count gates are $23, 29, 37$. These opposite trends ask whether sufficiently high redundancy has any sporadic fields outside the persistent and modular loci.

The structural conclusion is nevertheless stable. Beyond redundancy four, deepness is no longer controlled by one invariant or by isolated bad fibers. The persistent geometry is a coherent rank-two polar flag; modular geometry appears through characteristic-specific contraction kernels; and the remaining arithmetic is the Frobenius action on a normalized ordered-root cover. Redundancy five is the first exceptional-cover case, while redundancies six and seven show how that cover is propagated or contained.

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